Introduction to Data Science and Big Data

12th November 2024

Marking Scheme

Markers are advised that, unless a task specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) must be given the available marks. If there is doubt as to the correctness of an answer, the relevant NCC education materials should be the first authority.

This marking scheme has been prepared as a guide only to markers and there will frequently be many alternative responses which will provide a valid answer.

Each candidate's script must be fully annotated with the marker's comments (where applicable) and the marks allocated for each part of the tasks.

Throughout the marking, please credit any valid alternative point.

Marker's comments:		
Moderator's comments:		
Mark:	**Moderated mark:**	**Final mark:**
Penalties applied for academic malpractice:		

Marks

Question 1

a) Explain the key differences between data science, big data, **and** data analytics. **5**

Mark scheme:
1 mark for each correct point to a maximum of 5 marks

Data science is the overarching field that focuses on extracting insights and value from structured and unstructured data by employing techniques such as machine learning, statistics, and advanced analytics (1). Big data refers to the vast and complex datasets that are generated at high velocity from various sources, making them difficult to process with traditional data management tools (1). Data analytics, on the other hand, is the specific process of analysing data to discover patterns, trends, and insights that assist in decision-making (1). Data science encompasses both big data and data analytics, utilising a variety of tools, technologies, and methods to process and analyse data effectively (1). While big data primarily deals with challenges like volume, variety, velocity, and veracity, data analytics focuses on interpreting the data to derive meaningful conclusions (1). Typically, data analytics involves querying, reporting, and applying statistical methods to datasets, whereas data science integrates these techniques with algorithms to predict outcomes and automate decisions (1).

Any suitable alternative allowed.

b) Provide **ONE (1)** example where data science, big data **and** data analytics overlap in real-world applications. **5**

Mark scheme:
1 mark for each correct point to a maximum of 5 marks

In the healthcare industry (1), data science, big data, and data analytics often overlap in managing patient care. Big data is used to collect vast amounts of patient information from various sources (1), such as electronic health records, wearable devices, and genetic data (1). Data analytics is then applied to this data to uncover patterns and trends (1), such as identifying high-risk patients based on their medical history and lifestyle factors (1). Data science combines both by using machine learning algorithms to create predictive models (1) that forecast potential health outcomes (1), such as the likelihood of developing chronic diseases (1). These insights help healthcare providers personalise treatment plans (1), improve patient outcomes (1) and reduce hospital readmission rates (1). This integration of data science, big data, and data analytics enhances decision-making (1) and optimises the quality of healthcare services (1).

Any suitable alternative allowed.

Total 10 Marks

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Marks

Question 2

a) Define the 5Vs of big data **and** explain why each is critical in managing large datasets. **5**

Mark scheme:

1 mark for each bullet point with correct definition of v and the reason why it is critical in managing large datasets.

The 5Vs of big data are critical in managing large datasets.

- ***Volume refers to the massive amount of data generated from various sources, which requires scalable storage solutions (1).***
- ***Velocity represents the speed at which data is created and processed, often in real-time, demanding fast data processing technologies (1).***
- ***Variety involves the different types of data, such as structured, unstructured, and semi-structured data, which necessitates flexible data handling tools (1).***
- ***Veracity pertains to the accuracy and reliability of the data, ensuring high-quality data for analysis (1).***
- ***Lastly, Value refers to the insights derived from the data, making it essential for decision-making and driving business outcomes (1).***

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

b) Identify data privacy and security issues in big data applications. For each issue explain how they are controlled. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

Data privacy and security challenges in big data solutions are handled through several key approaches.

- ***Unauthorized access to sensitive information: This is controlled by data encryption, which secures data both at rest and in transit, ensuring that unauthorized parties cannot access the data. (1)***
- ***Unrestricted or inappropriate access to data: Controlled by access control mechanisms, such as role-based access control (RBAC), which limit who can view or modify specific data based on their role within the organization. (1)***
- ***Privacy breaches due to exposure of personally identifiable information: Mitigated by data anonymisation techniques, which remove or mask personally identifiable details from datasets, reducing the risk of privacy violations. (1)***
- ***Outdated security measures or non-compliance with regulations: Addressed through regular security audits and compliance protocols, such as adhering to standards like GDPR, to ensure that security measures remain current and effective. (1)***
- ***Data breaches or anomalies going undetected: Controlled by monitoring systems that continuously track data activity, allowing organizations to detect, respond to, and mitigate the impact of any breaches or suspicious behaviour. (1)***

Any suitable alternative allowed

Total 10 Marks

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Marks

Question 3

a) Define what is meant by the term **data**. **1**

Mark scheme:

Data refers to information that is collected, stored, and analysed to help make decisions, solve problems, or understand phenomena.

b) Identify **and** explain **THREE (3)** primary categories of data. **6**

Mark scheme:

1 mark for identification and 1 mark for explanation. Max 2 marks for each bullet point.

The main types of data are:

- ***Structured (1): Structured data is organised in a predefined format, typically in rows and columns, such as databases or spreadsheets (1)***
- ***Unstructured (1): Unstructured data lacks a defined structure and includes formats like text, images, and videos (1).***
- ***Semi-structured (1): Semi-structured data contains elements of both structured and unstructured data, such as JSON and XML files, where some organisation exists but is not in a rigid format (1).***

Any suitable alternative allowed.

c) Provide **ONE (1)** real-world scenario example of each data type you identified in (b). **3**

Mark scheme:

1 mark for each correct point to a maximum of 3 marks

- ***Structured data can be found in databases such as customer transaction records in a retail system (1).***
- ***Unstructured data is exemplified by social media posts, emails or videos, where the content lacks a specific format (1).***
- ***Semi-structured data is commonly seen in JSON or XML files, like logs from web servers or data shared between systems via APIs (1).***

Any suitable alternative allowed.

Total 10 Marks

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Marks

Question 4

a) Describe the concept of data preprocessing. Why is it important in the data science process? **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks. Both parts of the question should be answered correctly to be awarded 5 marks.

- ***Data preprocessing involves cleaning (1) and transforming raw data (1) into a usable format for analysis (1).***
- ***It includes steps like handling missing values (1), normalising data (1), and removing duplicates (1).***
- ***This process is crucial in the data science pipeline because raw data often contains errors or inconsistencies (1) that can affect model performance (1).***
- ***Proper preprocessing ensures that the data is clean, accurate, and in the correct format (1) which leads to more reliable and accurate models (1).***
- ***Ultimately improving the quality of insights drawn from the analysis (1).***

Any suitable alternative allowed.

b) Explain the key steps involved in data preprocessing. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

The key steps in data preprocessing include:

- ***Data cleaning which involves removing duplicates and handling missing values (1),***
- ***Data transformation where data is normalised or standardised for consistency (1),***
- ***Data integration combining data from multiple sources to create a unified dataset (1),***
- ***Data reduction which reduces the dimensionality by selecting important features (1),***

and data encoding converting categorical data into numerical form for model compatibility (1).

Any suitable alternative allowed.

Total 10 Marks

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Marks

Question 5

a) Discuss the differences between descriptive and inferential statistics, including examples of each along with their purposes. **6**

Mark scheme:

1 mark for each correct point to a maximum of 6 marks

Descriptive statistics summarise and describe the main features of a dataset (1) through measures such as mean, median, mode and standard deviation (1). They provide a clear overview of the data by highlighting central tendencies and variability (1).

Inferential statistics, on the other hand, use a sample of data to make predictions or generalisations about a larger population (1). Techniques like hypothesis testing, confidence intervals and regression analysis are used in inferential statistics (1). While descriptive statistics describe the data at hand, inferential statistics help draw conclusions and make decisions about future data or populations (1).

Any suitable alternative allowed.

b) Provide **ONE (1)** real-world example each of when descriptive **and** inferential statistics would be used in a data science project. **4**

Mark scheme:

1 mark for each correct point to a maximum of 2 marks for each example.

In a data science project, descriptive statistics might be used to summarise customer purchase behaviour by calculating the average spending, median age and standard deviation, helping to identify trends and outliers (1). It provides a clear picture of the customer base and highlights key patterns (1).

Inferential statistics would be used when predicting future sales based on a sample of historical data, such as using hypothesis testing to assess whether a new marketing strategy significantly increases sales (1). It could also involve calculating confidence intervals to predict future trends and testing whether findings from a sample can be generalised to a larger population (1).

Any suitable alternative allowed.

Total 10 Marks

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Marks

Question 6

a) Explain the concept of Exploratory Data Analysis (EDA). **3**

Mark scheme:

1 mark for each correct point to a maximum of 3 marks

- ***Exploratory Data Analysis (EDA) is the process of investigating datasets to summarise their main characteristics using both graphical and numerical methods (1).***
- ***EDA is crucial for understanding data structure, guiding further analysis and informing decision-making. (1)***
- ***It typically serves as the first step before formal modelling (1).***

Any suitable alternative allowed.

b) Explain graphical **and** numerical methods used in EDA, with **TWO (2)** examples each and discuss **ONE (1)** difference between them. **7**

Mark scheme:

1 mark for each correct point to a maximum of 7 marks

- ***Graphical methods of Exploratory Data Analysis (EDA) involve visualising data to identify patterns, trends and outliers (1).***
- ***Examples:***
  - ***Histograms, which show data distribution (1),***
  - ***Scatter plots which reveal relationships between variables (1),***

- ***Numerical methods of EDA focus on summarising data using statistics like mean, median and standard deviation to describe central tendency and variability (1).***
- ***Examples:***
  - ***Descriptive Statistic which includes measurement such as mean, medium and variance provides a numerical summary of data distribution (1).***
  - ***Frequency Tables summarise the number of occurrences of different values in a dataset (1)***
- ***While graphical methods help in quickly interpreting large datasets and provide a visual insight, numerical methods give precise, quantitative information (1).***

Any suitable alternative allowed.

Total 10 Marks

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Marks

Question 7

a) Explain in detail what is meant by ***feature engineering*** and discuss why it is important in machine learning. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

- ***Feature engineering is the process of transforming raw data into meaningful features that can be used by machine learning algorithms (1).***
- ***It involves creating new variables, selecting relevant features and encoding data to improve model performance (1).***
- ***Feature engineering is crucial because the quality and relevance of features directly affect a model's accuracy and ability to generalise (1).***
- ***It helps algorithms better capture patterns in the data, leading to more accurate predictions and better decision-making (1).***
- ***By improving data representation, feature engineering enhances the overall performance of machine learning models (1).***

Any suitable alternative allowed.

b) Give **ONE (1)** example of a feature engineering technique that improves model performance. Discuss the benefits of the technique you have identified. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

- ***One common feature engineering technique is one-hot encoding (1)***
- ***which is used to convert categorical data into a numerical format (1).***
- ***For example, in a dataset with a "Colour" feature containing categories like "Red," "Blue" and "Green," one-hot encoding creates separate binary columns for each colour (1).***
- ***This allows machine learning models to interpret categorical features numerically (1).***
- ***By representing categorical variables in a form the algorithm can understand, one-hot encoding helps improve model performance, especially in classification problems (1).***
- ***This technique reduces bias that might occur from assigning arbitrary numerical values to categories (1).***

Any suitable alternative allowed.

Total 10 Marks

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Marks

Question 8

a) Discuss the main differences between supervised **and** unsupervised learning. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

- ***Supervised learning involves training a model on labelled data, where the input and corresponding output are provided (1).***
- ***The model learns the relationship between input features and the target variable to make predictions, such as in classification or regression tasks (1).***
- ***In contrast, unsupervised learning deals with unlabelled data, where the model must identify patterns or structures on its own (1).***
- ***Common tasks in unsupervised learning include clustering and dimensionality reduction (1).***
- ***The key difference is that supervised learning requires labelled data and is used for prediction, while unsupervised learning seeks to find hidden patterns without predefined outputs (1).***

Any suitable alternative allowed.

b) Provide **ONE (1)** example of a problem where each type of learning is best suited. **5**

Mark Scheme:

1 mark for each correct point to a maximum of 5 marks

- ***Supervised learning is best suited for problems like spam detection in emails (1).***
- ***In this case, a model is trained on a labelled dataset where emails are marked as "spam" or "not spam" (1),***
- ***allowing the model to predict the label of new emails based on learned patterns (1).***
- ***Unsupervised learning is ideal for problems like customer segmentation in marketing (1).***
- ***Here, the model groups customers into segments based on similar purchasing behaviours or demographics without predefined labels (1),***
- ***helping businesses target different customer groups more effectively (1).***

Any suitable alternative allowed.

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Marks

Question 9

a) Discuss the role of Business Intelligence (BI) in modern organisations. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

- ***Business Intelligence (BI) plays a critical role in modern organisations by transforming raw data into actionable insights that guide decision-making (1).***
- ***BI tools help organisations monitor key performance indicators (KPIs) and track operational metrics in real-time (1),***
- ***allowing them to make data-driven decisions that enhance efficiency and productivity (1).***
- ***By providing visualisations like dashboards and reports, BI enables stakeholders to identify trends, forecast future outcomes, and respond to market changes quickly (1).***
- ***Additionally, BI aids in identifying areas for improvement, optimising processes, and gaining a competitive advantage in the industry (1).***

Any suitable alternative allowed.

b) Describe how BI tools can help in improving operational efficiency. Illustrate your answer with an example from any industry. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

- ***Business Intelligence (BI) tools improve operational efficiency by providing real-time data insights that help organisations make informed decisions quickly (1).***
- ***For example, in the retail industry, BI tools like dashboards and reports can track inventory levels, sales patterns and customer behaviour (1).***
- ***This allows retailers to optimise stock levels, reducing overstock or stockouts, and ensuring products are available when needed (1).***
- ***Additionally, by analysing sales trends, retailers can adjust pricing or promotions to maximise revenue (1).***
- ***BI tools streamline operations, reduce inefficiencies, and enable faster response to changing market conditions (1).***

Any suitable alternative allowed.

Total 10 Marks

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Marks

Question 10

a) Discuss the purpose and significance of ethics in data science. Illustrate your answer with examples. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

- ***The significance of ethics in data science lies in ensuring the responsible use of data to protect individuals' rights and promote fairness (1).***
- ***Ethical practices help prevent misuse of sensitive information, such as personal or financial data, ensuring privacy and confidentiality (1).***
- ***Ethics also address the risk of bias in data collection and algorithmic decision-making, which can lead to unfair outcomes or discrimination (1).***
- ***Additionally, transparency and accountability are critical to maintaining public trust in data-driven technologies (1).***
- ***Overall, ethical considerations are essential for promoting fairness, protecting privacy, and ensuring responsible use of data in decision-making (1).***

Any suitable alternative allowed.

b) Explain FIVE (5) ways in which organisations ensure ethical use of data in their projects. **5**

Mark scheme:

1 mark for each correct point to a maximum of 5 marks

- ***Organisations can ensure the ethical use of data by implementing clear data governance policies (1),***
- ***complying with regulations like GDPR (1),***
- ***anonymising personal data to protect privacy (1),***
- ***conducting bias assessments on algorithms to ensure fairness (1),***
- ***providing transparency in data handling practices (1),***
- ***and regularly training employees on ethical data use (1).***
- ***Regular audits and monitoring of data practices further ensure ethical compliance (1).***

Any suitable alternative allowed.

Total 10 Marks

End of paper

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Learning Outcomes matrix

Questions	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO 1	Yes
2	LO 1, 2	Yes
3	LO 2, 3	Yes
4	LO 2, 3	Yes
5	LO 2, 4	Yes
6	LO 1, 2, 4	Yes
7	LO 4	Yes
8	LO 1, 4	Yes
9	LO 1,4, 5	Yes
10	LO 1, 5, 6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Demonstrate a systematic and thorough understanding of the concept of Data Science and Big Data	*Exhibits minimal grasp of foundational concepts in Data Science and Big Data.* *Fails to showcase an ability to identify, adapt, or utilise appropriate skills, methods, and procedures effectively in addressing data-related challenges.*	*Displays an insufficient understanding of essential concepts in Data Science and Big Data.* *Struggles to articulate and apply appropriate skills, methods, and procedures adequately, indicating a need for further development.*	*Achieves a satisfactory level of understanding in the fundamental concepts of Data Science and Big Data.* *Demonstrates the ability to identify, adapt, and apply foundational skills, methods, and procedures to address standard data challenges.*	*Attains a very good understanding of advanced concepts in Data Science and Big Data.* *Exhibits a proficient ability to identify, adapt, and effectively use a range of skills, methods, and procedures, surpassing basic requirements.*	*Attains an outstanding level of comprehension in intricate concepts of Data Science and Big Data.* *Demonstrates an exceptional ability to identify, adapt, and apply a comprehensive set of skills, methods, and procedures, yielding well-supported, highly nuanced solutions that showcase depth and mastery.*

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Demonstrate proficiency in data collection, design, and modelling for data processing	*Struggles to demonstrate basic proficiency in data collection, design, and modelling for data processing.* *Exhibits limited understanding of fundamental concepts and is unable to apply them effectively.*	*Displays some proficiency in data collection, design, and modelling for data processing but falls short of the expected standard.* *Shows inconsistencies or weaknesses in applying relevant skills, methods, and procedures.*	*Demonstrates a satisfactory level of proficiency in data collection, design, and modelling for data processing.* *Applies foundational skills, methods, and procedures effectively, achieving competence in standard scenarios.*	*Attains a very good level of proficiency in data collection, design, and modelling for data processing.* *Demonstrates a well-developed ability to apply a range of skills, methods, and procedures effectively, surpassing basic requirements.*	*Achieves an exceptional level of proficiency in data collection, design, and modelling for data processing.* *Displays mastery in applying a comprehensive set of advanced skills, methods, and procedures, showcasing depth and excellence in addressing complex data processing challenges.*
Select appropriate techniques and tools for data pre-processing.	*Struggles to select or apply basic techniques and tools for data pre-processing.*	*Shows some ability to select techniques and tools but falls short of the expected standard.*	*Demonstrates a satisfactory ability to select appropriate techniques and tools for data pre-processing.*	*Attains a very good ability to select and apply a range of techniques and tools for data pre-processing.*	*Achieves an exceptional ability to select and apply advanced techniques and tools for data pre-processing.*
Understand the analytical techniques and software tools to effectively generate insights from data.	*Struggles to select or apply basic techniques and tools for data pre-processing.*	*Shows some ability to select techniques and tools but falls short of the expected standard.*	*Demonstrates a satisfactory ability to select appropriate techniques and tools for data pre-processing.*	*Attains a very good ability to select and apply a range of techniques and tools for data pre-processing.*	*Achieves an exceptional ability to select and apply advanced techniques and tools for data pre-processing.*
Understand the suitability of different visualisation methods for different types of Data	*Struggles to understand the basic suitability of visualisation methods for different types of data.*	*Displays some understanding of the suitability of visualisation methods but falls short of the expected standard.*	*Demonstrates a satisfactory understanding of the suitability of visualisation methods for different types of data.*	*Attains a very good understanding of a range of visualisation methods and their suitability for different data types.*	*Achieves an exceptional understanding of advanced visualisation methods and their nuanced suitability for different data types.*

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Demonstrate a critical understanding of Data Science Ethics	*Struggles to demonstrate a basic understanding of data science ethics.*	*Displays some understanding of data science ethics but falls short of a critical analysis.*	*Demonstrates a satisfactory understanding of data science ethics.*	*Attains a very good understanding of a range of ethical considerations in data science.*	*Achieves an exceptional understanding of advanced ethical considerations in data science.*

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